

	Republic of the Philippines Department of Transportation
	Document No: <u>DOTr-HRDD-Forms-004</u> Rev. No.: <u>001</u> Effective Date: <u>03 August 2022</u>
	RE-ENTRY ACTION PLAN (REAP) FORM

INSTRUCTIONS: This form is used in preparing a proposed Re-entry Action Plan in applying to a Scholarship Program. Duly accomplished and signed forms will be submitted to the HRD Division along with the other documentary requirements.

Title of Scholarship Program:	JDS 23 rd Batch Scholarship Program
Name of the Development Partner	Japan International Cooperation Center (JICE) through the National Economic Development Authority (NEDA)

I. REAP BACKGROUND AND INFORMATION	
a. REAP Title	Advancing Maintenance Competencies in Railway through Development of a Systematic Framework and Evaluation Criteria for the Philippine Railways Institute (PRI)
b. REAP Objective	The applicant proposes to recommend or create a systematic maintenance inspection workflow and a designed evaluation criteria ideal for the local and existing maintenance procedures. It will also integrate international standards, advanced technologies, and innovations gained from the scholarship program. Through the conduct of Capacity Development Training (CDT) course offered by the Philippine Railways Institute (PRI), applicant plans to demonstrate a maintenance competency-based topic that discusses the proposed evaluation framework aiding railway personnel to improve their maintenance inspection procedures in their respective workplace. The objective is to develop an evaluation framework for maintenance inspection, specifically a risk-based maintenance prioritization frameworks that employs a six-level evaluation (AA to S) to classify defects based on urgency and severity. Then to operationalize this framework, the applicant plans to present it to the organization's office and proposes to integrate it into a new maintenance-focused training module through the Capacity Development Training (CDT) course offered by the Philippine Railways Institute (PRI). The module will aim to enhance the competencies of railway personnel by improving maintenance inspection practices and promoting safer, more reliable operations. To present the framework to the organization for adoption, then within the next CDT course revision cycle, preparations will be made for training sessions to outline the designed evaluation framework, along with the development of training materials.
c. Problem/Opportunity	Metro Manila's railway system faces challenges such as aging infrastructure, high passenger volumes, and limited maintenance resources. One key factor that the applicant observes is that it

necessitates a more systematic approach to prioritizing maintenance activities. The applicant aims to address this gap by proposing an evaluation framework to guide railway personnel and explore how various strategies and new insights gained from the scholarship program can be adapted to Metro Manila's maintenance activities prioritization.

As described in the objective, the evaluation framework for maintenance inspection provides a structured approach that could inform the development of risk-based maintenance prioritization frameworks that composes of a six-level evaluation (AA to S) to classify defects based on urgency and severity. In this framework, the prioritization of repair is based on classification of the defects. Immediate repairs are prioritized for "AA" classifications, addressing safety-critical issues, while "A1" issues are addressed within a year. Lower-level classifications ("A2," "B," and "C") are monitored for progression.

The proposed initiative aligns with the Philippine Development Plan (PDP) 2023–2028 by contributing to enhanced infrastructure planning, programming, and asset management. Specifically, it supports the goal of upgrading and expanding infrastructure (Chapter 12), which emphasizes asset management as key to ensuring efficient, safe, and sustainable operations. Furthermore, the initiative addresses the PDP's focus on human capital development (Chapters 2 and 4) through capacity-building activities aimed at enhancing the competencies of railway personnel. Additionally, through this initiative, this will align with the Sustainable Development Goals (SDGs) Goal 9: Industry, Innovation, and Infrastructure, fostering resilient infrastructure through systematic maintenance prioritization and defect management, thereby reducing risks and enhancing reliability and promoting innovative maintenance strategies by incorporating advanced methodologies and insights from international standards into local practices. The project also creates opportunities for skills development and ensures safer working environments for railway personnel as the purpose for Goal 8: Decent Work and Economic Growth in Sustainable Development Goals (SDGs).

The Philippine Railways Institute, a research and training center under the administrative supervision of the DOTr, shall serve as the planning, implementing, regulating, and administrative entity for the development of human resources in the railway sector to ensure efficient, reliable, and safe railway transportation services. One of its main functions is carried out by the Training Division, responsible for providing training to all existing and new railway personnel. Additionally, it is responsible for the conduct and update of training needs assessment of the railway sector by formulating, developing, and publishing curricula, training materials, textbooks, and instructors' manuals involving the operations and maintenance of railway to upgrade the competencies of railway personnel. One of the Training Courses offered by Philippine Railways Institute (PRI) is the Capacity Development Training, it is a training conducted to update the railway operations and maintenance personnel who are already certified by the Institute after five (5) years on new laws, regulations, and technologies in the railway sector. Generally, it is planned as a one (1)-day training course and it will be delivered through a combination



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	of various teaching methodologies such as lectures, case studies, group discussions, and hands-on practice. The objective of the training course is to provide railway personnel with the necessary knowledge, skills, and attitude to sustain the effective and safe operation and maintenance of the railway system. This gives the applicant an opportunity to present the proposed evaluation framework through the Capacity Development Training (CDT) course offered by the Philippine Railways Institute to railway personnel and which can be a recommending initiative that aims to improve their maintenance inspection procedures in their respective workplace thus, strengthening their competencies, ensuring they remain capable of meeting the demands of a growing and modernizing railway sector.
d. Description in relation to the Department i. Organizational Outcome	The primary outcome of this Re Entry Action Plan (REAP) is to develop a risk-based maintenance prioritization framework for railway infrastructure, utilizing a six-level evaluation scale (AA to S) to classify defects based on urgency and severity. This framework, combined with its integration into a maintenance-focused training module through the Capacity Development Training (CDT) course at the Philippine Railways Institute (PRI), will address challenges in service delivery and workforce competency. <u>Enhancing Service Delivery</u> The implementation of a systematic maintenance prioritization framework will significantly improve the reliability and safety of Metro Manila's railway systems. Classifying defects based on urgency and severity, the framework ensures that maintenance resources are allocated efficiently, addressing critical issues and reducing risks of service interruptions. This proactive approach to maintenance will enhance the quality of railway services, leading to safer and more reliable transportation for passengers. Furthermore, incorporating international standards and advanced methodologies into local practices ensures that the railway system adopts best practices in asset management, aligning with the Philippine Development Plan (PDP) 2023-2028's emphasis on enhancing infrastructure planning and asset management (Chapter 12). This contributes to the development of resilient infrastructure and promotes sustainable service delivery. <u>Capacity Development</u> The Re Entry Action Plan (REAP) will introduce proposed framework the into the maintenance competency-based topic within the Capacity

	Development Training (CDT) course, ensuring personnel have ideas and knowledge to the latest methodologies, standard, and innovative tools used in international railway systems. It will directly address the need for capacity-building among railway personnel by integrating the framework into a new module for the CDT course. This ideally aligns with the PDP's focus on human capital development (Chapters 2 and 4), fostering a competent workforce that can support Metro Manila's growing transportation needs. Lastly, the REAP contributes to the PRI's mandate of delivering efficient, safe, and reliable railway services. It also supports long-term goals by promoting a culture of continuous learning and innovation within the organization. Ultimately, the REAP ensures sustainable improvements in infrastructure maintenance, which benefits both the organization and the public it serves.
ii. Target Competency Improvement	The program at the applicant's interested University department aligns well with the competencies needed to execute his Re-Entry Action Plan (REAP). Specifically in Infrastructure Management and Transportation Planning, it will equip the applicant with the technical expertise to design, implement, and manage infrastructure systems, including risk-based maintenance prioritization. The program's focus on infrastructure management, transportation planning, urban development, and international infrastructure standards will enhance his ability to integrate innovative maintenance strategies into local practices, addressing Metro Manila's railway challenges. Courses also on data analytics and decision-making in terms of transportation planning, will equip the applicant with the ability to classify defects effectively and utilize resource allocation using real-time data. These skills are important for addressing the limitations of current maintenance systems. Understanding and modeling predictive defects into data of interest using innovative technologies are currently available at the program of study which will aid in the framework development. Additionally, exposure to innovative technologies and facilities available will enhance the applicant's ability to incorporate and share these innovative solutions into the local railway infrastructure development, which are important for sustainability and improving service delivery. <u>Contribution of the Program of Study to the Re Entry Action Plan:</u> The scholarship program's focus on infrastructure management, human resource development, and innovative technologies aligns directly with the goals of the applicant's Re Entry Action Plan. This will provide the applicant with the tools and guidelines needed to the development of his REAP. By applying the knowledge gained from the scholarship program, the applicant will contribute to the professional growth of railway personnel and the overall safety and efficiency of the railway system.
e. Relation to the Government's Development Plan	The REAP, which aims to enhance Metro Manila's railway maintenance through a risk-based prioritization framework, aligns directly with the goals set in the Philippine Development Plan (PDP) 2023-2028. Specifically, it supports the goal of upgrading and expanding infrastructure (Chapter 12), focusing on improving asset management to ensure efficient, safe, and sustainable operations. This contributes to the resilient infrastructure needed to accommodate the



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	growing demands of urban transportation, as emphasized in the Philippine Development Plan (PDP). Furthermore, the Philippine Development Plan (PDP) highlights the importance of continuous workforce development to sustain infrastructure facilities and operational effectiveness. The REAP addresses the human capital development goals (Chapters 2 and 4), ensuring that railway personnel have the skills required to implement modern maintenance strategies and improve service delivery. Addressing the need for enhanced competencies in railway maintenance personnel, the REAP directly supports the government's commitment to improving transportation systems as a cornerstone of national development. The Re-Entry Action Plan (REAP) aligns with the United Nations Sustainable Development Goals (SDGs) specifically Goal 9: Industry, Innovation, and Infrastructure, fostering resilient infrastructure through the adoption of systematic maintenance prioritization, advanced defect management, and international best practices. It also provides skills development opportunities through structured training programs, enhancing the competencies of railway personnel as align with Goal 8: Decent Work and Economic Growth.
f. Direct Beneficiary	The implementation of this REAP will provide immediate advantages to certified railway operations and maintenance (O&M) personnel within the Philippine Railways Institute (PRI). Particularly those directly involved in maintenance and inspection tasks. These personnel, who have undergone initial training and certification, are the primary beneficiaries, as the proposed framework for evaluation criteria in maintenance inspection can serve as a reference in their current workplace. Technically, they can also gain knowledge and skills as this will align with international standards, guidelines, and advanced maintenance technologies. Additionally, the proposed risk-based maintenance prioritization framework which will discussed through training sessions in Capacity Development Training (CDT) course can improve capacity of railway personnel by introducing new maintenance methodologies that can be related to common strategies, such as predictive maintenance from using non-destructive testing, participants will gain practical knowledge to improve their efficiency in diagnosing and addressing railway maintenance issues. The training will also emphasize updated international railway standards, enabling personnel to utilize railway maintenance inspections.

	Indirect beneficiaries on the other hand, include railway passengers, who will experience improved service reliability and safety due to better-maintained infrastructure and facilities. In the long term, the railway sector benefit from a more competent workforce and enhanced operational efficiency. This Re Entry Action Plan (REAP) not only strengthens the capacity development of railway personnel but also contributes to the long goal of delivering a safer, efficient, and reliable railway transportation system for the public.
g. Gender Equality, Disability, and Social Inclusion (GEDSI)	The implementation of this Re Entry Action Plan (REAP) will actively promote inclusivity and equitable opportunities for women and other marginalized groups, including Indigenous Peoples (IP), Persons with Disabilities (PWD), and members of the LGBTQ++ community, in the railway sector, a traditionally male-dominated roles such as railway maintenance. Integrating strategies and methodologies that address their unique needs, the Re Entry Action Plan (REAP) ensures that the developed risk-based maintenance framework and enhanced Capacity Development Training (CDT) course is accessible for all participants. <u>Strategies for Inclusivity:</u> Gender-Responsive Content: Training materials will include examples and case studies that are applicable to women and marginalized groups in railway operations and maintenance, promoting their representation and inspiring others. Additionally, understanding and learning how they handle and promote respectful and inclusive communication in this scholarship program and share these learnings through the institute and in the delivery of training course. Accessibility: The training venue, materials, and delivery methods developed under the REAP will be designed to accommodate PWD participants, including the use of accessible facilities. This ensures equal opportunities for all railway personnel, regardless of physical ability. Encouraging Participation: Efforts will prioritize in encouraging women and marginalized groups in the CDT course, ensuring they have equal access to professional development opportunities. The innovative learnings that the applicant can gain on how the accessibility of women and marginalized groups is such as persons with disabilities (PWDs) and indigenous people (Ips) in their own railway transportation could also indirectly benefit them by implementing it thru these new learnings.
h. Output	The Re Entry Action Plan (REAP) will generate several outputs that contribute directly to achieving the desired organizational outcomes: <u>Risk-Based Maintenance Prioritization Framework:</u> A systematic and structured framework for classifying and prioritizing maintenance tasks based on severity and urgency of defects in railway infrastructure. This can be a recommendation to improve asset management, utilize resources, and enhance operational efficiency within the country's railway transportation system.



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Training Module for Railway Personnel: A comprehensive training program designed to enhance the Capacity Development Training (CDT) course which is aimed at improving the competencies of railway maintenance personnel, including technical skills, the use of new methodologies, and international practices. This will ensure that personnel are equipped to implement the new maintenance framework and contribute to a more reliable and efficient system.

Comprehensive Training Materials: Detailed handouts and trainee guides for the new training topic. These materials will align with international best practices while addressing the specific needs of railway sector.

Quality Training Session: Delivery of training session of the proposed topic under the Capacity Development Training (CDT) course, ensuring the recommended topic's practical application and knowledge sharing to railway personnel.

Standards Recommendation: Suggestions for integrating the developed framework for evaluation criteria which references new insights and learnings relating to railway standards can be used as references from various local railway operators.

II. REAP TIMELINE AND IMPLEMENTATION

a. Measurability

It is imperative to establish a clear milestones and key performance indicators (KPIs) to monitor the success of the applicant's REAP implementation. Generally, it aligns with the objectives of improving maintenance processes and are listed into the following:

Development and Approval of Risk-Based Framework: Completion of the prioritization framework for maintenance defects, with internal approval and integration into the capacity development training (CDT) course offered by the Philippine Railways Institute (PRI).

Completion of Training Module: Development and delivery of a comprehensive training program for railway personnel.

Assessment and Feedback: Conduct evaluations and gather feedback from trainees after training sessions and initial implementation.

Additionally, these outcomes will enhance the CDT course by integrating modern advancements and technologies into the training course and providing measurable indicators of success such as the training evaluation which is conducted after the course. A satisfactory rating results from the training evaluation validates the relevance of the developed training topic and corresponding training module.

Training Evaluation Results: Post-training feedback from participants will be analyzed to assess the effectiveness, relevance, and impact of the training. Satisfactory ratings in the evaluation will indicate alignment with learning outcomes.

Milestones

After Scholarship Program ended and returning to work:

- Preparation of a draft of outcomes and new learnings from the scholarship program.

1 month after scholarship program:

- Share new learnings and outputs through an echo session with DOTR PRI-Training Division members
- Preparations for the development of proposed risk-based maintenance prioritization framework

Discussion and consultation of the proposed risk-based maintenance prioritization framework

- Conduct discussion and seek consultation from supervisors of the developed risk-based maintenance prioritization framework to check its applicability in local maintenance procedures.

After discussion and approval of the proposed risk-based maintenance prioritization framework

- Preparations for the inclusion of the developed risk-based maintenance prioritization framework into the topic in Capacity Development Training Course (CDT).

After CDT Course Topic Approval:

- Preparation and submission of training materials and modules of the approved CDT course topic integrating the knowledge, advanced technologies, and innovations gained from the scholarship program within 3 weeks after topic approval.

Capacity Development Training Course (CDT) training modules and materials revision period:

- Inclusion of the developed risk-based maintenance prioritization framework into the topic in Capacity Development Training Course (CDT).

During CDT Course

- Conduct training session of the approved Capacity Development Training Course (CDT) training topic.

After CDT Course

- Receive feedback and achieve an improvement or a satisfactory rating from the CSAT and Training Evaluation from the CDT Course Batches and use it as an indicator to check that railway personnel benefited from the conducted training course.

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b. Sustainability			Knowledge and learnings gained from this training can be institutionalized and sustained beyond a certain timeframe by deliberate planning and strategies. This includes documentation and knowledge transfer by creating a comprehensive documentation outlining the training objectives, methodologies, and outcomes. This can also serve as a reference for future instructors or colleagues, ensuring continuity in training practices. Transfer of knowledge can also be done through proposed echo session.
c. Predictive Action Steps with Identified Success Indicators			
i. Phase/Stage	ii. Expected Output	iii. Timeline	
After Scholarship Program ended and returning to work	Draft of outcomes and new learnings from the scholarship program.	2028	
1 month after scholarship program:	New learnings and outputs are shared on an echo session conducted with DOTR PRI-Training Division members	2028	
During discussion and consultation of the proposed risk-based maintenance prioritization framework	Preparations for the development of proposed risk-based maintenance prioritization framework with consultation from supervisors	2028	
After discussion and approval of the proposed risk-based maintenance prioritization framework	Development of proposed maintenance-focused training course topic in Capacity Development Training Course (CDT).	2028	
	Preparations for the inclusion of the developed risk-based maintenance prioritization framework into the topic in Capacity Development Training Course (CDT).		

Capacity Development Training Course (CDT) training modules and materials revision period:	Inclusion of the developed risk-based maintenance prioritization framework into the topic in Capacity Development Training Course (CDT).	2028
3 weeks after CDT Course Topic Approval:	Developed appropriate training materials and modules of the approved CDT course topic integrating the knowledge, advanced technologies, and innovations gained from the scholarship program	2028
During CDT Course	Training session of the developed Capacity Development Training Course (CDT) training topic is conducted.	2028
After CDT Course	Feedback is received a satisfactory rating from the CSAT and Training Evaluation from the CDT Course Batches is achieved which serves as an indicator to check that railway personnel benefited from the conducted training course	2028
d. Target Date to Start	2028 After Scholarship Program ended and returning to work	
e. Target Date to End	2028 After approval of the proposed risk-based maintenance prioritization framework and after the conduct of training delivery of the developed topic in the CDT course training	
III. RESOURCES		
The resources to be used in this proposed REAP are laptops and internet connection which will aid the instructor to prepare and list the new learnings and outputs into a presentation for the section head's approval to be added in the CDT course training materials. Projectors will also be used during the presentation of the revised training materials.		
All these items are available property of the DOTR-PRI		
IV. COMMENT(S) AND RECOMMENDATION(S)		
a. Immediate Superior (Mentor)		

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b. Head of the Office/ Division/Service	
c. Human Resource Development Division	

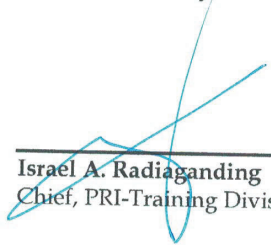
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Jeremy S. Regino
OIC-Undersecretary and Executive Director
Pursuant SO No. 2024-431